COMSATS University Islamabad
Department of Management Sciences · BS (Business Data Analytics)

Configuration Control Dashboard
for any configurable product

One live view of every part, version and approved change

Student photo

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Project Goals

1. Track every item and variant
2. Approve every change
3. Detect drift automatically
4. Audit-ready in seconds
5. Work for any product type

Works for Any Product

- Smartphones
- Electric cars
- Aircraft
- Medical devices
- Industrial machines

Proven on a real case study: **Google Pixel 7** — **194** items · **73** real builds · **12** variants · **6** drift rules · **104** tests

What is the CCD?

A web dashboard that shows **what is inside every product variant** and raises an alarm when anything **changes without approval**.

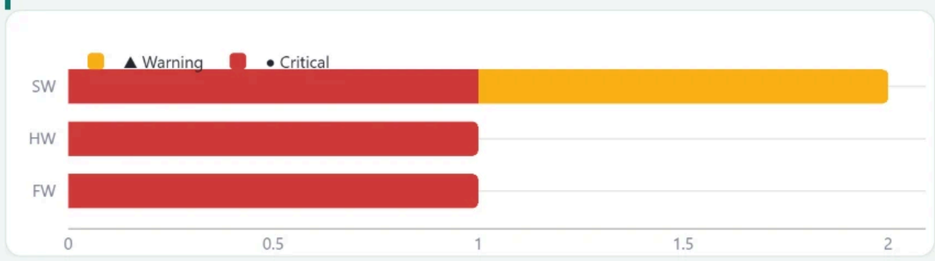
- Register of all parts
- Approved baselines
- Change requests (Jira)
- Drift alerts
- Compatibility check
- One-click audit report

Overview

Is the Pixel 7 configuration healthy right now? Every number links to the page that lists it.

CIs	Active baseline	Open ECRs	● Critical	▲ Warning	Compliance
194	MB-2026-09	0	3	1	97.8%

Open findings by CI category



severity	rule_type	ci_id	status	variant	actual_version
Critical	unauthorized_change	HW-011	Unauthorized	all	E
Critical	unauthorized_change	SW-001	Unauthorized	G03Z5-JP-128	CP2A.260705.006
Critical	unauthorized_change	FW-002	Unauthorized	all	g5300q-260604-260807-b-1610000
Warning	variant_inconsistency	SW-001	Unauthorized	G03Z5-JP-128	CP2A.260705.006

Sync status per source

source_system	status	records_in	changed
github	✓ OK	8	14
history	✓ OK	8094	9397
jira	✓ OK	35	242
plm_mock	✓ OK	1103	13

Register by category

category	CIs
DC	25
FW	17

Case study (Pixel 7): unapproved firmware change caught instantly **3 Critical** **1 Warning**

How It Works

APPROVED baseline + **ACTUAL** deployed state + **AUTHORIZED** change requests

✓ Match: compliant

● Mismatch: alarm

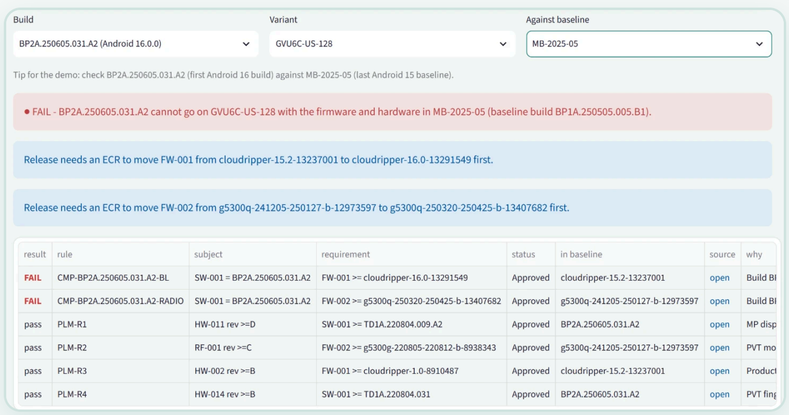
Software repo (GitHub) · Change requests (Jira) · PLM / BOMs · Release data

→ Sync → Database → Engine → Dashboard

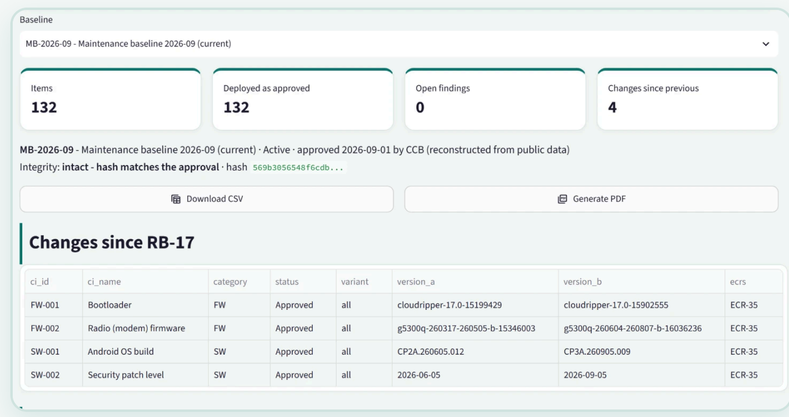
Inside the Dashboard



Variant Matrix red = differs



Compatibility blocks misfits



Audit Report one-click PDF

Project Plan

1. Research
2. Data & Sync
3. Engine
4. Dashboard
5. Testing

✓ All phases complete

Deliverables

- ✓ 8-page dashboard
- ✓ Live data sync
- ✓ Drift detection
- ✓ Audit reports

Built With

Python · Streamlit · FastAPI · SQLAlchemy · Plotly · GitHub API · Jira API